

BF26.5

1. Using the right hand rule, you can determine the angle the magnetic field must be. For the negative charge, you must use the opposite of the negative charge.

2. Knowns: $\vec{B} = 0.18 \text{ T}$, $d = 50 \sqrt{2} \text{ cm}$

Unknowns: $\int_i^f \vec{B} \cdot d\vec{s} = ?$

$$\begin{aligned}\int_i^f \vec{B} \cdot d\vec{s} &= \int_{0 \text{ m}}^{0.5 \cdot \sqrt{2} \text{ m}} 0.18 \text{ T } dx \\ &= (0.18 \text{ T})(0.70711 \text{ m}) \\ &= \boxed{0.127 \text{ T} \cdot \text{m}}\end{aligned}$$

3. Knowns: $\vec{B} = 0.18 \text{ T}$, $d = 50 \sqrt{2} \text{ cm}$

Unknowns: $\int_i^f \vec{B} \cdot d\vec{s} = ?$

Since the magnetic field is always perpendicular to the integration path, the dot product of field with direction will always be 0, meaning $\int_i^f \vec{B} \cdot d\vec{s} = 0 \text{ T} \cdot \text{m}$.

4. Knowns: $I_1 = 2 \text{ A}$, $I_2 = 6 \text{ A}$, $\oint \vec{B} \cdot d\vec{s} = 5.03 \cdot 10^{-6} \text{ T} \cdot \text{m}$

Unknowns: $I_3 = ?$

$$\begin{aligned}\oint \vec{B} \cdot d\vec{s} &= \mu_0 I_{enc} \\ I_{enc} &= \frac{5.03 \cdot 10^{-6} \text{ T} \cdot \text{m}}{1.26 \cdot 10^{-6} \text{ T} \cdot \text{m} / \text{A}} \\ &= 4 \text{ A} \\ I_3 &= I_{enc} - (I_1 + I_2) \\ &= 4 \text{ A} - (-6 \text{ A} + 2 \text{ A}) \\ &= \boxed{8 \text{ A}}\end{aligned}$$

5. Knowns: $I_2 = -12 \text{ A}$, $\oint \vec{B} \cdot d\vec{s} = 1.72 \cdot 10^{-6} \text{ T} \cdot \text{m}$

Unknowns: $I_3 \text{ direction} = ?$, $I_3 = ?$

Since I_2 goes out of the page, it's magnetic field is counterclockwise. Since the path is clockwise and the magnetic field integral is positive, the net magnetic field should also be clockwise. Thus, I_3 should go into the page.

$$\begin{aligned}
 I_{enc} &= \frac{\oint \vec{B} \cdot d\vec{s}}{\mu_0} \\
 &= \frac{1.72 \cdot 10^{-5} \text{ T} \cdot \text{m}}{1.26 \cdot 10^{-6} \text{ T} \cdot \text{m} / \text{A}} \\
 I_{enc} &= 13.7 \text{ A} \\
 I_2 &= 13.7 \text{ A} - (-12 \text{ A}) \\
 &= \boxed{26 \text{ A}}
 \end{aligned}$$

6. Knowns: $q = 1.6 \cdot 10^{-19} \text{ C}$, $v = 8 \cdot 10^6 \text{ m/s}$, $B = 0.46 \text{ T}$

Unknowns: $\vec{F}_1 = ?$, $\vec{F}_2 = ?$

$$\begin{aligned}
 |\vec{F}_1| &= q\vec{v} \times \vec{B} \\
 &= (1.6 \cdot 10^{-19} \text{ C})(8 \cdot 10^6 \text{ m/s})(0.46 \text{ T}) \sin(45^\circ) \\
 &= 4.16 \cdot 10^{-13} \text{ N}
 \end{aligned}$$

$\vec{F}_1$ points in the direction of positive y by the right hand rule

$$\vec{F}_1 = 4.16 \cdot 10^{-13} \text{ N } \hat{j}$$

$$\begin{aligned}
 |\vec{F}_2| &= q\vec{v} \times \vec{B} \\
 &= (1.6 \cdot 10^{-19} \text{ C})(8 \cdot 10^6 \text{ m/s})(0.46 \text{ T}) \sin(90^\circ) \\
 &= 5.89 \cdot 10^{-13} \text{ N}
 \end{aligned}$$

$\vec{F}_2$ points in the direction of positive z by the right hand rule

$$\vec{F}_2 = \boxed{5.89 \cdot 10^{-13} \text{ N } \hat{k}}$$

7. Knowns: atomic masses, $\vec{B} = 3T$

Unknowns: $f_{O_2^+} = ?$, $f_{N_2^+} = ?$, $f_{CO^+} = ?$

$$\begin{aligned}
 F_c &= F_b \\
 \frac{mv^2}{r} &= qvB \\
 v &= \frac{qBr}{m} \\
 f &= \frac{v}{2\pi r} \\
 f &= \frac{qB}{2\pi m} \\
 f_{O_2^+} &= \frac{(1.6022 \cdot 10^{-19} \text{ C})(3.0000 \text{ T})}{2\pi(2(15.995)(1.6605 \cdot 10^{-27} \text{ kg}))} \\
 &= \boxed{1.4401 \text{ MHz}} \\
 f_{N_2^+} &= \frac{(1.6022 \cdot 10^{-19} \text{ C})(3.0000 \text{ T})}{2\pi(2(14.003)(1.6605 \cdot 10^{-27} \text{ kg}))} \\
 &= \boxed{1.645 \text{ MHz}} \\
 f_{CO^+} &= \frac{(1.6022 \cdot 10^{-19} \text{ C})(3.0000 \text{ T})}{2\pi((12.000 + 15.995)(1.6605 \cdot 10^{-27} \text{ kg}))} \\
 &= \boxed{1.6457 \text{ MHz}}
 \end{aligned}$$

8. Knowns: $v = 0.1c$, $r = 0.22 \text{ m}$, $q = 1.6 \cdot 10^{-19} \text{ C}$

Unknowns: $B = ?$

$$\begin{aligned}
 v &= \frac{qBr}{m} \\
 3 \cdot 10^7 \text{ m/s} &= \frac{(1.6 \cdot 10^{-19} \text{ C})B(0.22 \text{ m})}{1.67 \cdot 10^{-27} \text{ kg}} \\
 B &= 1.42 \text{ T}
 \end{aligned}$$

9. Knowns: $B = 0.41 \text{ T}$, $v = 0.6 \cdot 10^7 \text{ m/s}$, $q = -1.6 \cdot 10^{-19} \text{ C}$

Unknowns: $\vec{F}_1 = ?$, $\vec{F}_2 = ?$

$$\begin{aligned}
 \vec{F} &= q\vec{v} \times \vec{B} \\
 |\vec{F}_1| &= (1.6 \cdot 10^{-19} \text{ C})(0.6 \cdot 10^7 \text{ m/s})(0.41 \text{ T}) \sin(90^\circ) \\
 &= 3.936 \cdot 10^{-13} \text{ N} \\
 \vec{F}_1 &\text{ points in the negative z direction by the opposite of the right hand rule} \\
 \vec{F}_1 &= \boxed{-3.94 \cdot 10^{-13} \text{ N } \hat{k}} \\
 |\vec{F}_2| &= (1.6 \cdot 10^{-19} \text{ C})(0.6 \cdot 10^7 \text{ m/s})(0.41 \text{ T}) \sin(90^\circ) \\
 &= 3.936 \cdot 10^{-13} \text{ N} \\
 \vec{F}_2 &\text{ points in a } 45^\circ \text{ angle to the y axis by the opposite of the right hand rule} \\
 \vec{F}_2 &= \boxed{-2.78 \cdot 10^{-13} \text{ N } \hat{j} - 2.78 \cdot 10^{-13} \text{ N } \hat{k}}
 \end{aligned}$$